Started on	Tuesday, 18 May 2021, 8:16 PM
State	Finished
Completed on	Tuesday, 18 May 2021, 9:40 PM
Time taken	1 hour 24 mins
Grade	**40.00** out of 50.00 (**80**%)

Question 1

Partially correct

Mark 7.00 out of 10.00

Ahmad does a part-time job as a cashier at JayaGrocery since last year. He obtained an average wage of RM10.50 an hour with a standard deviation of RM1.50. If the wages are approximately normally distributed and paid to the nearest cent.

(Note: 2 dp = 2 decimal places)

(a) What percentage of Ahmad receive wages between RM9.50 and RM12.0 an hour inclusive?

The z values corresponding to X = 9.50 is

-0.67

✔ (2 dp)

The z values corresponding to X = 12.0 is

1.0

✔ (1 dp)

Percentage =

59.0

✔ % (1 dp)

(b) The highest 5% of the hourly wages are greater than what amount?

P(Z = ?) = 1 -

0.05

✔ =

0.95

✔ (2 dp)

X =

1.65

✖ (1 dp)

Question 2

Partially correct

Mark 6.00 out of 10.00

The administrators of a Faculty A were interested to understand personal characteristics of their students. They decided to conduct a survey and managed to get randomly 1625 students to respond to the survey. The same survey has been conducted for 3 years in a row and a summary report on students' characteristics was produced.

The first part of the survey was asking demographics questions on the students. The data in Table 1 show eight variables, each describing a different characteristic of the students taking the survey. For the data in the table, the description of the data is as follows:

ID : An identifier for each of the individuals in the study.

Gender : M for male and F for female

Mathematics : To measure how much the student likes Mathematics courses (*Really Don't Like it, Not too much, I like it, Absolutely love it*)

Exercise : Number of times spent exercising per week

Games : Number of hours spent playing online games per week

GPA : Current grade point average on a 4-point scale

Pulse : Pulse rate in number of beats per minute at the time of the survey

Table 1: Sample of Result of student survey

ID	Gender	Mathematics	Exercise	Games	GPA	Pulse
1	M	Not too much	3-4	1	3.13	54
2	M	Not too much	More than 6 times	7	2.5	66
3	M	Really Don't Like it	0	5	2.55	130
...						
...						
1623	M	I like it	1-2	8	3.3	77
1624	F	Not too much	1-2	6	2.8	60
1625	F	Absolutely love it	0	1	3.7	94

1. Based on the case study, what is the :

1. population:

Students in Faculty A ✔

2. primary data

Student data from the 3 years' survey ✖

2. Identify the data type and its level of measurement for each data below:

Data	Data type	Level of Measurement
Exercise	Discrete ✔	Ratio ✖
Games	Continuous ✔	Interval ✖
Pulse	Discrete ✖	Ratio ✔

3. This year, Faculty A decided to add a second part of the survey. They were interested in the student's opinions on the new café in the faculty building. Identify if the possible answers are Qualitative or Quantitative Data:

	Questions	Qualitative or Quantitative Data
i.	Why do think students like the new café?	Qualitative ✔
ii.	What is the suitable price for drinks served at the café?	Quantitative ✔

Question 3

Partially correct

Mark 5.00 out of 7.00

(a) *Student A would like to visit a country where 60% of the population speaks the native language.*

However, Student A does not speak and understand that language.

What is the probability that at most one of five locals at the immigrant department of the airport will be able to communicate with Student A?

b(0;5,0.40)+b(1;5,0.40) ✔

= 0.33696 ✔

(b) *Research Scientist A investigates the infected zebrafish in modeling human diseases.*

Research Scientist A finds out the probability of the zebrafish getting infected is three out of thirty.

What is the probability that ten zebrafish must be required before Research Scientist A finds at least eight zebrafish but not all have been infected?

b*(10;8,0.10)+b*(10;9,0.10) ✔

= 0.0000002997 ✔

(c) *A baseball game consists of three bases, and each base consists of nine innings.*

A batter against whom three strikes are recorded strike-out.

The probability of Player A and Player B earning a hit in their batters are 90% and 85%.

What is the probability difference between Player A and Player B to get their first strikes in the second inning?

Player A: g(2;0.10) ✖

Player B: g(2;0.15) ✖

Difference = 0.0375 ✔

Question 4

Correct

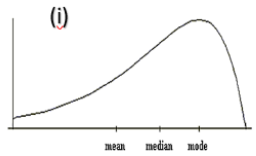
Mark 3.00 out of 3.00

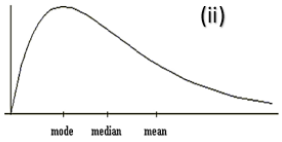
The duration of wearing the *kodenshi far-infrared* socks for a group of patients who claimed that the socks re-energized their fatigue was organized into a frequency distribution. The mean, median, and mode duration are 3.3 hours. The standard deviation was computed to be 0.42 hours.

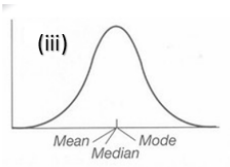
(a) The distribution is: Symmetry ✔

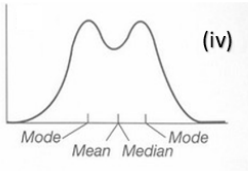
(b) Justify your answer in (a): Mean,Median,Mode is equal ✔

(c) Which of the following graph represent the dispersion of central tendency of your answer in (a):









Answer: iii ✔

Question 5

Correct

Mark 10.00 out of 10.00

Mrs. Linda distributes numbers of cotton balls to each of her students. Her students received the cotton balls based on the given below:

18	13	26	12	20	14	21	16	17
22	17	25	13	30	13	22	15	21
18	18	16	21	18	31	15	19	10

a) Construct a **stem-and-leaf** plot for the given dataset.

Stem	Leaf
1 ✔	0 2 3 3 3 4 5 5 6 6 7 7 8 8 8 9 ✔
2	0 1 1 1 2 2 5 6 ✔
3	0 1

Key: 1|0 means 10 ✔

b) Find the following of cotton balls received for each student.

i. Range

31 ✔ -

10 ✔ =

21 ✔

ii. Mode

18 ✔

iii. Mean **(up to 2 decimal points)**

501 ✔ / 27 =

18.56 ✔

Question 6

Partially correct

Mark 4.00 out of 5.00

Table following shows the CGPA of a class of 15 students who undergoes the semester with fully online classes.

2.95	2.99	3.04	3.04	3.05	3.22	3.24	3.25	3.34	3.35	3.35	3.52	3.64	4.00	4.00

(a) Calculate the following:

(i) 40th percentile = 3.23 ✔

(ii) The percentile for 3.64 = 86 ✖

(iii) Q2 = 3.25 ✔

(iii) Inter Quartile Range = 0.48 ✔

(b)Which measurement will be the best suitable in justifying the performance of the students for this case? Why?

MEAN, because it shows the overall achievement and no outliers in the data ✔

Question 7

Correct

Mark 5.00 out of 5.00

The table below shows the height measurement collected from students who are finishing primary school. Using the information provided in the table, answer the following questions.

Height, h	Frequency, f
156.5 – 159.5	5
159.5 – 162.5	13
162.5 – 165.5	12
165.5 – 168.5	10
168.5 – 171.5	21
171.5 – 174.5	9

(a) Fill in the following table:

Height, h	Frequency, f	Mid Point,x	fx
156.5 – 159.5	5	158	790 ✔
159.5 – 162.5	13	161	2093
162.5 – 165.5	12	164	1968 ✔
165.5 – 168.5	10	167 ✔	1670
168.5 – 171.5	21	170	3570 ✔
171.5 – 174.5	9	173	1557 ✔

(b) Calculate the **MEDIAN** (*provide the answer in 2 decimal places.*)

Median class = 165.5 – 168.5 ✔

Median = 167.00 ✔

(c) Calculate the **MEAN** (*provide the answer in 2 decimal places.*)

mean = $\sum f_i, x_i$ / 70

= 11648 ✔ / 70

= 166.40 ✔

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